

# Structured grounding layer — design markup (top pick, before build)

Parse the manuals' own RPSTL parts tables & maintenance work packages into structured records → exact part auto-fill, look-alike warnings, and a verbatim procedure panel. Every field stays cited to its page.

|                                |                      |                  |                          |                             |                     |                    |
|--------------------------------|----------------------|------------------|--------------------------|-----------------------------|---------------------|--------------------|
| In your corpus (sample index): | ITEM NO <b>4,223</b> | SMR <b>4,590</b> | PART NUMBER <b>3,764</b> | USABLE ON CODE <b>3,748</b> | torque <b>1,561</b> | lb-ft <b>1,030</b> |
|--------------------------------|----------------------|------------------|--------------------------|-----------------------------|---------------------|--------------------|

**1 · EXTRACTION (offline, idempotent, resumable)**

**RPSTL / -24P parts pages**  
Parse repeating rows: FIG · ITEM · SMR · CAGEC · NSN · PART NUMBER · NOMENCLATURE · USABLE-ON-CODE.

**Maintenance work packages (WP)**  
Parse tasks: REMOVAL / INSTALLATION / DISASSEMBLY → numbered steps, tools, torque values — captured verbatim with page refs.

↓

**→ schema migration 0004 (additive, R1) fills the reserved tables:**  
parts (+ cagec, smr, fig\_no, item\_no, uoc, nomenclature, doc, page, vehicle) · part\_variants (differs\_how, how\_to\_tell) · procedures (kind, steps, tools, torque, doc, page) · figures.  
  
Confidence score per record; the source page is always kept for citation.

**A · Exact part auto-fill**  
Add a part → NSN, part#, FIG, nomenclature fill from the structured RPSTL row (not a guessed snippet), cited to the catalog page.

**B · Look-alike warning**  
Same nomenclature, different NSN/part#? Show '⚠ 2 variants — check Usable-On-Code' with the cited distinguishing column. Only when the data shows a real difference.

**C · Verbatim procedure panel**  
For a part/fault: remove/install steps, tools, and torque shown VERBATIM with TM+page — never paraphrased. Your original goal C.

+ OCR coverage meter (parallel): /api/coverage → per-vehicle '% searchable' shown in the hub & search.

## 3 · SEQUENCING

**PHASE 1 (foundation)**  
Structured RPSTL parts + look-alike variants + exact auto-fill. Migration 0004. Coverage meter. Quick wins: multi-sheet pagination (>6 items) + capture confirm/override.

build first

**PHASE 2 (verbatim procedures)**  
Procedure panel: remove/install/tools/torque, cited & verbatim, linked from a part or fault. Builds on Phase-1 parts & figures.

next

**PARALLEL (OCR)**  
Keep completing GPU OCR; each pass raises extraction coverage for both phases. Coverage meter makes the progress visible.

ongoing

**GROUNDING GUARANTEE**  
Extraction is OCR/table-noisy, so: nothing is shown without its cited source page; low-confidence records are flagged for the mechanic to verify; a look-alike difference is only stated when the manual's data actually shows it. We surface the manual — we never paraphrase specs or invent a variant.